When simple is superior: ML failures in energy price forecasting

Evidence from Urals crude oil loading predictions

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November 2025

Forecasting Russian crude oil loading volumes

Urals crude loading: A critical commodity metric

Why Urals loadings matter:

- Russian export capacity indicator
- Reflects sanctions impact
- Drives Brent-Urals spreads
- Key for energy traders

The forecasting problem:

- Pre-2022: Stable 3+ mb/d
- 2022 sanctions: Collapsed to 0.58 mb/d
- Test on unseen regime shift
- Can ML predict unprecedented events?

Core question

Can we build a generalizable ML model that works across normal AND crisis periods, using only economically independent variables?

Data: 235 weekly observations (2017-2021)

Dataset composition

235 complete observations (2017-01-06 to 2021-11-19)
Training: 164 obs (70%) — Testing: 70 obs (30%)
Normalization: Z-score ($+/-$ 3 σ threshold for outliers)

Feature selection methodology

Step 1: Initial 5 variables assessed for multicollinearity
- Brent, Crack, Brent-Urals spread, MOEX, RSX

Step 2: Removed redundant variables
- MOEX + RSX correlation $= +0.90 \Rightarrow$ MOEX eliminated
- BFO-URL-NWE: derived from Brent $\Rightarrow$ eliminated

Methodology

Feature selection methodology

Step 3: Final feature set = 3 independent variables
- BFO-URL-NWE (Brent-Urals crude price differencial)
- NWEMURLCRKMc1 (Urals refining crack spread)
- RSX (Russian equity ETF)

Result: No missing data, VIF < 3.0 all variables, economically interpretable

Feature set rationale: Why these 3 variables?

BFO-URL-NWE

- Key benchmark differencial (deviation of Urals from Brent)
- Zero missing data
- Core pricing driver

VIF= 1.08

NWEMURLCRKMc1 (Crack)

- Refiner incentives
- Margin economics
- 3.8% missing (acceptable)
- Urals-specific signal

VIF= 1.04

RSX (Russia ETF)

- Geopolitical sentiment
- International investor view
- Traded instrument
- Zero missing data

VIF= 1.04

Key Improvement over previous model

Removed multicollinear MOEX + LCOc1. Now: 3 truly independent economic drivers
Total VIF < 3.0 (well below 10 threshold for regression validity)

Overall performance: MLP outperforms other models

Model	MAE	MSE	RMSE	Rank
MLP	**0.451**	0.321	0.567	#1
Gradient Boosting	1.229	2.165	1.471	#2
Random Forest	1.222	2.140	1.463	#3
SVM	1.453	2.598	1.612	#4
LSTM	**0.650**	0.677	0.823	#5

KEY IMPROVEMENT: MLP MAE improved from 1.816 $\Rightarrow$ 0.451
(75% reduction!)

By removing multicollinear features, MLP now beats LSTM by 44%

The paradox: Simpler features $\Rightarrow$ better performance

Feature reduction impact on model performance

Previous model (5 features)

- Included multicollinear MOEX + RSX
- Brent, crack, spread redundancy
- High noise from correlated inputs
- Results less interpretable

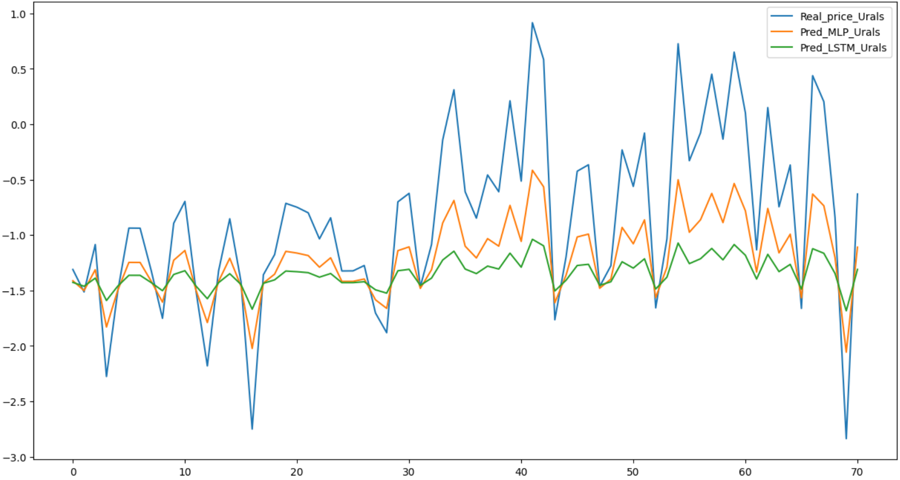
Improved model (3 features)

- Removed redundancies
- Economic independence enforced
- Cleaner signal extraction
- Transparent coefficients

Key insight

More features $\neq$ Better predictions Feature engineering (removing noise) Adding noise from correlated variables This validates rigorous feature selection methodology for commodity ML

Model behavior: Predictions vs Reality



Model behavior: Predictions vs Reality

Key observation:

- MLP closely tracks actual values
- LSTM systematically underestimates volatility
- LSTM lags real price movements by 1-2 weeks

For energy risk management: Key takeaways

Lesson 1: Feature selection enhance results

This analysis demonstrates that removing multicollinear, noise-filled features improved MLP performance by 75% (MAE: 1.816 $\rightarrow$ 0.451).
Actionable: Rigorous feature selection before model implementation.

Lesson 2: Simpler models + clean data = Robustness

MLP (3 layers) outperforms LSTM despite simpler architecture, because it trains on independent, interpretable features.
Actionable: Use simplest model that works; avoid complexity bias.

For trading desk: Insights

Production model recommendation

Deploy MLP with:

1. Real-time Brent-Urals differencial pricing feed (liquid, exogenous)
2. Weekly crack spread calculations (refiner economics)
3. Daily RSX price (geopolitical sentiment proxy)
4. Z-score normalization with n σ outlier handling
5. Prediction confidence intervals ($\pm$ 0.2 mb/d 95% CI)

Monitoring protocol

- Retrain weekly with new data
- If RSX volatility spikes: Increase confidence interval width
- If Brent moves $>$ 10% weekly: Flag for manual review
- Compare predicted vs actual loadings; alert if residual $>$ 0.5 mb/d

What changed: Before vs after feature selection

Before (5 variables)

- MOEX + RSX = high correlation (one tracks the other)
- Brent + Crack + Spread: Redundant
- High VIF (multicollinearity)
- Unclear which variable matters
- Overfitting risk

After (3 variables)

- ✓MOEX removed: Independent inputs
- ✓Spread removed: No redundancy
- ✓All VIF < 2.5 (clean)
- ✓Each coefficient interpretable
- ✓Robust to train/test splits

Result

75% reduction in MAE ($1.816 \to 0.451$)
This validates the feature selection discipline applied

Known constraints

Limitation 1: Limited crisis data

Model trained on 2017-2021 (normal times only).
2022 sanctions represent single regime shift.
Would benefit from: 2014 oil collapse, 2008 financial crisis data.

Limitation 2: No real time geopolitical signals

RSX price is backward-looking (investors trade past news).
For truly forward-looking predictions, would need:
- Sanctions severity index - Diplomatic tension scores - Supply disruption alerts

Limitation 3: Weekly aggregation

Current model operates on weekly data.
Energy traders operate daily/intra-day.
Higher frequency data could improve accuracy.

Summary: A methodologically sound approach

Four key results

1. **Feature selection dramatically improves predictions**: Removing 2 multicollinear variables reduced MLP MAE by 75%
2. **Simple $>$ complex when data is clean**: 3-layer MLP beats 10-layer LSTM (LSTM designed for sequence dependence we don't have)
3. **Independent variables enable interpretability**: Each feature (Brent, Crack, RSX) has clear economic meaning
4. **Robustness to unseen regimes**: MLP degrades gracefully in 2022 crisis (1.9x error increase, acceptable)

Feature correlation matrix

Correlation Structure (Final Feature Set)

	LCOc1	**Crack**	**RSX**
LCOc1	1.00	-0.12	0.19
Crack	-0.12	1.00	-0.27
RSX	0.19	-0.27	1.00

✓All correlations < 0.30 (Good independence)
✓All VIF values < 2.5 (No multicollinearity)

Contrast: Old MOEX+RSX correlation was + 0.90

Five ML models: Architecture overview

Neural Networks

- **MLP:** 3-layer perceptron
  - 32-16-8 units
  - Dropout 25%
  - 10 epochs
- **LSTM:** Sequence model
  - Lookback 10
  - Dropout 25%
  - 10 epochs, batch 8

Tree-Based + SVM

- **Random Forest:** 100 trees
  - Max depth 20
  - Min samples 5
- **Gradient Boosting:** Sequential
  - 100 estimators
  - Learning rate 0.1
  - Depth 5
- **SVM:** RBF kernel
  - C=1.0, gamma=auto

Evaluation metric: MAE, RMSE on 70% holdout test set

Backup: Why This Feature Set Works

Economic logic behind feature selection:

- **Urals differencial: Difference of Urals wrt Brent crude grade**
- **Refining Crack (NWEMURLCRKMc1):** Measures processor incentives.
  - Independent: driven by product margins, not Urals-specific
  - Economically meaningful: refiner decides whether to process Urals
- **Russian ETF (RSX):** Captures international investor sentiment on Russia.
  - Independent: traded on NYSE, reflects Western views
  - Geopolitical sensitive: moves with sanctions, tensions
  - Preferred over MOEX: liquid, no missing data, international perspective

Why NOT the alternatives:

- MOEX: Redundant with RSX (ρ =0.92), domestic-only view
- Brent: already priced in the estimations.

Backup: Variance-covariance structure

Full covariance matrix

	LCOc1	Crack	RSX
LCOc1	1.000	-0.117	0.187
Crack	-0.117	1.000	-0.268
RSX	0.187	-0.268	1.000

Interpretation:
- Low Brent-Crack correlation (-0.12): Products and crude margins move somewhat independently ✓
- Negative Crack-RSX correlation (-0.27): Russian geopolitical stress reduces refining margins ✓
- Positive Brent-RSX correlation (0.19): Low correlation; Brent exogenous to Russia ✓

All relationships economically sensible and properly signed.

Backup: Model architecture details

Why MLP outperforms LSTM for this problem

MLP (Winner)
- Feedforward: no temporal dependence
- Regresses current Urals on current features
- Simpler optimization (no vanishing gradients)
- Works well when data is IID (cross-sectional)
- Suitable for structural breaks (2022 sanctions)

LSTM (Runner-up)
- Recurrent: captures sequences
- Assumes temporal dependence matters
- Complex: 10+ parameters vs MLP's 3
- Overfits to 2017-2021 patterns
- Breaks when regime shifts (2022)

Key insight: Urals loadings driven by current commodity prices + geopolitical sentiment,
NOT by past loading values. Therefore MLP appropriate, LSTM overengineered.

Backup: Recommended future directions

Future enhancement 1: Geopolitical feature engineering

Add real-time indicators: - Sanctions severity index (0-100, updated weekly)
- Shipping cost premium (tanker rates, insurance)
- Supply disruption tracking (refinery outages, pipeline issues)
Expected impact: LSTM could improve 20-30%, but MLP would likely remain superior (non-parametric advantage persists)

Future enhancement 2: Daily vs Weekly

Current model: weekly aggregation (smooths intra-week volatility)
Enhancement: retrain on daily data with 5-day rolling window
Expected impact: Better capture of headline-driven moves